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Dominance of ST131, B2, *bla*$_{CTX-M-15}$, and *papA-papC-kpsMII-uitA* among ESBL *Escherichia coli* isolated from bloodstream infections in Quito, Ecuador: a 10-year surveillance study (2009–2019)

Jeannete Zurita[1,2,3,*], Gabriela Sevillano[2], Ariane Paz y Miño[2], Nathalí Haro[2], Marco Larrea-Álvarez[4], Iliana Alcocer[5], David Ortega-Paredes[4,6,7]

[1]Facultad de Medicina, Pontificia Universidad Católica del Ecuador, Quito 170525, Ecuador
[2]Unidad de Investigaciones en Biomedicina, Zurita & Zurita Laboratorios, Quito 170104, Ecuador
[3]Servicio de Microbiología y Tuberculosis, Hospital Vozandes, Quito 170521, Ecuador
[4]Facultad de Ciencias Médicas Enrique Ortega Moreira, Carrera de Medicina, Universidad Espíritu Santo, Km 2.5 vía a Samborondón 0901952, Ecuador
[5]Escuela de Ciencias Biológicas, Pontificia Universidad Católica del Ecuador, Quito 170525, Ecuador
[6]Unidad de Investigación en Enfermedades Transmitidas por Alimentos y Resistencia a los, Antimicrobianos (UNIETAR), Facultad de Veterinaria, Universidad Central del Ecuador, Quito 170129, Ecuador
[7]Laboratorio de Referencia de *E. coli*, Department of Microbiology and Parasitology, Veterinary Faculty, University of Santiago de Compostela, Lugo 15782, Spain

*Corresponding author. Unidad de Investigaciones en Biomedicina. Zurita & Zurita Laboratorios. 170104. E-mail: jzurita@zuritalaboratorios.com

Abstract

Aims: This study aimed to examine antibiotic resistance and the epidemiology of extended-spectrum β-lactamases (ESBL)-producing *Escherichia coli* associated with bloodstream infections over a period of 10 years.

Methods and results: Isolates were collected from January 2009 to December 2019 and those testing for *E. coli* were included. Antibiotic susceptibility was tested using the VITEK® system. Selected isolates were further characterized by amplification of marker genes (virulence traits, phylogroups, and sequence types). A total of 166 ESBL-producing *E. coli* were recovered. The *bla*$_{CTX-M-15}$ allele was the most abundant. Most of the isolates were resistant to ceftriaxone, cefepime, ceftazidime, ampicillin/sulbactam, piperacillin/tazobactam, and ciprofloxacin. No resistance to carbapenems was registered. More than 80% of bacteria were classified as extraintestinal pathogenic *E. coli* (ExPEC), and the combination of virulence traits: *papA-papC-kpsMII-uitA* was the most common. Phylogroup B2 was the most prevalent, and bacteria predominantly belonged to ST131.

Conclusions: There was an increase in the ExPEC ESBL-*E coli* in bloodstream infections and the relationship between the isolates found in these infections during these 10 years.

Impact Statement

Constant surveillance of antibiotic resistance among *Escherichia coli* bloodstream infections appears crucial not only for infection control but also for the implementation of effective therapies

Keywords: *E. coli*, BSI, ST131, *bla*$_{CTXM-15}$, surveillance

Introduction

Escherichia coli is considered one of the most common causes of bloodstream infections (BSIs) worldwide (Paterson and Bonomo 2005). Various reports have shown the relevance of *E. coli* not only for BSI but also in terms of contamination of animal products, the environment, and communities (Pitout et al. 2005). Antibiotic treatment for *E. coli* BSI could be compromised due to increasing antibiotic resistance; therefore, treatment options are diminishing (Tumbarello et al. 2010) Studies have observed that most *E. coli* BSI is associated with community onset and preceded by a urinary tract infection (UTI), mainly in elderly people, although other infectious sources could represent primary foci, such as intra-abdominal infections, including hepatic-biliary tract, respiratory tract, and soft tissue infections (Rodríguez-Baño et al. 2008, Vihta et al. 2018). Generally, extended-spectrum β-lactamases (ESBL) *E. coli* exhibits *in vitro* co-resistance to fluoroquinolones, aminoglycosides, and trimethoprim-sulfamethoxazole, and represents a considerable global public health concern (Kumar et al. 2014, Castanheira et al. 2021).

Although detailed investigations on the molecular epidemiology are key for developing preventive strategies, limited information is available on the prevalence of the bacteria responsible for BSI in Ecuador (Zurita et al. 2019). ESBL-producing pathogens are now commonplace in both hospital and community settings. Enterobacteriaceae producing-ESBL, such as SHV and TEM types, have been established

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since the 1980s as a major cause of hospital-acquired infections. However, since the 2000s, detection rates of CTX-M type ESBL-producing *E. coli* have been identified predominantly from the community as a cause of UTIs and in clinical isolates globally (Pitout et al. 2005). Additionally, ESBL-producing *E. coli* has been identified in urban rivers, companion and livestock animals, poultry products, vegetables, and ready-to-eat food (Ortega-Paredes et al. 2018a, 2019, 2020; Vinueza et al. 2019, Albán et al. 2020, Zurita et al. 2020, Salinas et al. 2021). A possible factor for the increased prevalence of ESBL *E. coli* is the global dissemination of high risk clones that frequently carry $bla_{\text{CTX-M-15}}$, such as ST131, ST44, ST10, or ST410 (Mathers et al. 2015). The aforementioned sequence types (STs) have been identified in specimens originating both from BSI and urban environmental niches (Ortega-Paredes et al. 2018a, 2020, Zurita et al. 2019). CTX-M β-lactamases are widespread enzymes, and early variants of CTX-M enzymes could hydrolyze ceftriaxone (CRO) and cefotaxime, although later variants could also hydrolyze ceftazidime (CAZ) (e.g. CTX-M-15 and CTX-M-55) (Castanheira et al. 2021). Although treatments are available for isolates producing these enzymes, the combination with other resistance mechanisms may limit the activity of new agents (Castanheira et al. 2021). Additionally, many of these new treatment alternatives are not currently available in Latin America (Terreni et al. 2021).

To understand the epidemiology of *E. coli* BSI, we performed a surveillance study of ESBL bacteria in a hospital in Quito, Ecuador, over a period of 10 years (2009–2019). Polymerase chain reaction (PCR) assays were performed to identify $bla_{\text{CTX-M}}$ variants, and the susceptibility of isolates to commonly used antibiotics was assessed. Moreover, key genes were identified to determine the virulence traits, phylogroups, and STs.

Materials and methods

Study population

This retrospective investigation included ESBL-producing *E. coli* isolates recovered from blood cultures at a local institution in Quito, Ecuador. Vozandes Hospital is a private hospital in Quito (68 beds). Isolates were obtained from patients with BSI from January 2009 to December 2019, and those testing for *E. coli* were included in this study. All etiological agents of BSI were maintained at the hospital's microbiology department in a cryopreserved collection. The samples are consecutive and nonduplicate BSIs. All phenotypic detection of ESBL production was confirmed by the double-disk synergy method that relies on inhibition with clavulanate (Jarlier et al. 1988).

Reactivation was performed by plating cultures in pre-enrichment medium (Tryptic Soy Broth) at 35°C for 24 h, followed by selective cultivation on ESBL-CHROMagar™ under the same conditions. Data, including antimicrobial susceptibility profiles, were retrieved from WHONET Software (https://whonet.org/).

Susceptibility to antibiotics

Isolates were recovered from positive samples detected by the BACTEC™ system (Becton Dickinson, USA) according to the guidelines. The standardized method for species identification and antibiotic resistance testing was implemented using the VITEK®2 system with GN, ID, and AST cards (bioMérieux, USA), following the manufacturer's recommendations. The tested antibiotics were as follows: ampicillin/sulbactam (SAM), piperacillin/tazobactam (TPZ), CRO, CAZ, cefepime (FEP), ertapenem (ETP), meropenem (MEM), imipenem (IPM), cefoxitin (FOX), ciprofloxacin (CIP), gentamicin (GEN), amikacin (AMK), and tigecycline (TGC). VITEK®2 Advanced Expert System software was used to interpret the results. For each year, the corresponding CLSI guidelines were used (CLSI 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019). For maintenance, isolates were frozen in skim milk (10%) with 15% glycerol at −80°C. MEM, IPM, and ETP were grouped as carbapenems.

Amplification of $bla_{CTX\text{-}M}$ genes

The boiling method was used for DNA extraction (Ahmed and Dablool 2017). PCR was performed on isolates having an ESBL resistance phenotype according to the VITEK®2 system. Identification of $bla_{\text{CTX-M}}$ genes (PCR conditions and primers) was carried out as described previously (Hasman et al. 2005). Subgroups 1, 2, 8, and 9 of $bla_{\text{CTX-M}}$ were identified as detailed previously (Hopkins et al. 2006, Jiang et al. 2006, Carattoli et al. 2008). Additionally, isolates were tested by PCR for the presence of the *mcr*-1 gene (Ortega-Paredes et al. 2016). The amplification products were confirmed by gel electrophoresis using a 1% agarose gel. All PCR products were sequenced at Macrogen Inc. (Seoul, South Korea). The obtained sequences were aligned against reference sequences using the online tool ResFinder 2.1, with an identity threshold of 100% (Ortega-Paredes et al. 2016). Confirmation of the variants was manually performed using Geneious Prime and the sequences retrieved from the Resfinder database (Ortega-Paredes et al. 2016).

Characterization of virulence traits

All ESBL-producing *E. coli* isolates were analyzed by PCR for virulence markers that define the extraintestinal pathogenic *E. coli* (ExPEC) status. An isolate was designated ExPEC when it was positive for at least two out of five determinants, *papA* and/or *papC*, *sfa*/*focDE*, *afa*/*draBC*, *iutA*, and *kps*MII (Johnson et al. 2003).

Determination of phylogenetic groups

Phylogenetic groups (A, B1, B2, C, D, E, and F) were determined using PCR by amplifying the *chuA*, *yjaA*, *arpA*, and *trpA* genes along with *TspE4.C2* fragment based on a published methodology (Clermont et al. 2013).

FumC/*fimH* typing

The approach is based on one of the household genes (*fumC*) used in a multilocus sequence typing scheme and a gene (*fimH*) encoding the internal fragment of a fimbrial adhesin, and the procedure was performed as previously described (Weissman et al. 2012). PCR amplification was carried out with appropriate primers, and products were analyzed on 2% agarose gels and then confirmed by sequencing (Macrogen Inc., Seoul, South Korea). The CHTyper web tool was used to identify specific CH types and establish the corresponding to specific STs using the EnteroBase database (http://enterobase.warwick.ac.uk) (Roer et al. 2018), and a minimal spanning tree was constructed with GrapeTree software (Zhou et al. 2018).

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Statistical analysis

Data were summarized using descriptive statistics. Fisher's test was used to assess differences in percentages. Shapiro–Wilk's and Levene's tests were utilized to determine the normality and homogeneity of variance, respectively. As the data proved to be non-normally distributed and heteroscedastic, a non-parametric test was used. The Spearman coefficient was calculated to determine tendencies between the number of isolates and the analyzed characteristics (antibiotic resistance, phylogroup, sequence type, ExPEC status, virulence traits, and $bla_{\text{CTX-M}}$ genes). Statistical significance was set at $P < 0.05$. Analyses were carried out in MATLAB® version 9.9.9341360 (MathWorks, Natick, MA, USA) (R2016a).

Ethical considerations

This research was carried out on frozen bacterial strains of *E. coli* from patients whose data were evaluated and stored anonymously. The Pontificia Catholic University of Ecuador waived the need for approval by a Committee on Bioethics in Human Beings and, as it was retrospective, individual informed consent was not obtained (Reference number: QINV0213-IINV5330100100).

Results

Study population

The total number of *E. coli* isolated between January 2009 and December 2019 from samples originating from BSI patients was 574. The numbers of total and characteristics of ESBL-producing *E. coli* isolated from bloodstream infections in the Vozandes Hospital, Quito-Ecuador per year are shown in Table 1. A total of 166 ESBL-producing *E. coli* isolates were obtained (Supplementary S1 Table 1). The mean age of the patients was 66.32 years, and 57.8% of the patients were female.

Characterization of $bla_{CTX\text{-}M}$ genes

The $bla_{\text{CTX-M-15}}$ variant was most commonly identified among isolates (67%), and the rest were present in <6.6% of isolates. Overall, the presence of the $bla_{\text{CTX-M-15}}$ variant increased during the studied period (Supplementary S2 Table 1).

Susceptibility to antibiotics

Over the course of 10 years, from 2009 to 2019, the number of ESBL-producing *E. coli* resistant to ampicillin-sulbactam (SAM), CRO, CIP, and GEN tended to increase (Supplementary S2 Table 1). Resistance to SAM, CRO, CIP, and GEN was lower in 2009 (7.5%, 3.8%, 7.5%, and 7.5%, respectively), and a progressive increase in resistance values was observed. Ten years later, in 2019, the resistance values of SAM, CRO, CIP, and GEN were 23.4%, 15.6%, 37.5%, and 26.6%, respectively. The highest resistance values were observed in 2014 (38.0%, 32.0%, 38.0%, and 40.0%, respectively). No significant increase in TPZ, CAZ, FEP, or FOX was observed over these years (Supplementary S2 Table 1). One isolate (2010) was resistant to AMK, and one isolate carried the *mcr*-1 gene. All isolates were susceptible to TGC and carbapenems (ETP, MEM, and IPM) (Supplementary S1 Table 1).

Characterization of virulence traits

In total, 82.5% of isolates were classified as ExPEC. An increasing trend in the number of ESBL-producing *E. coli* classified as ExPEC was detected (Supplementary S2 Table 1). Different combinations of virulence traits were found. In particular, *papA-papC-kpsMII-uitA* proved to be the most common (50%), followed by *iutA* (12%). The rest was detected in <5.4% (Supplementary S2 Table 1). Neither increasing nor decreasing trends were detected (Supplementary S2 Table 1). Only two isolates did not harbor the *kpsMII* or *iutA* genes.

Determination of phylogenetic groups

Amplification of marker genes allowed for the determination of phylogenetic groups. B2 was the most prevalent (68.1%), followed by A (15.7%), D (6.6%), B1 (4.8%), F (4.2%), and E (0.6%) (Table 1). Neither increasing nor decreasing trends were observed (Supplementary S2 Table 1).

FumC/fimH typing

Amplification of a household gene along with a fimbrial adhesin-encoding gene allowed for the assessment of CH types, and their corresponding specific STs types among isolates (Supplementary S1 Table 1). A total of 26 different STs were detected, and most of the bacteria belonged to ST131 (62%), followed by ST44 (7.8%) and ST38 (4.8%). The remaining STs were associated with fewer than seven isolates (4.2%) (Table 1). The analysis showed an association between ST131 and ST1193 and ST95. ST10 could be recognized as a potential ancestor of the ST44 group, whereas ST88 could be considered the origin of a variety of clones ST393, ST1706, ST69, ST2962, ST453, ST101, ST1128, ST5281, and ST602 (Fig. 1).

All isolates categorized as ST131 belonged to phylogroup B2 and were classified as ExPEC (Supplementary S1 Table 1), with the majority (76%) bearing the most common virulence pattern (*papA/C*, *kpsMII*, and *uitA*). Ninety-two percent of isolates carried the $bla_{\text{CTX-M-15}}$ allele, although some had other versions, including $bla_{\text{CTX-M-27}}$ and $bla_{\text{CTX-M-170}}$ (Table 1).

Over the course of 10 years, an increasing trend in the number of ESBL-producing *E. coli* classified as ST131 was detected (Supplementary S2 Table 1). These isolates have been registered since 2010 (Table 1). Moreover, all *E. coli* ST44 were classified as being part of phylogroup A, with 73% of them considered non-ExPEC. ExPEC isolates only contained two traits (*papC* and *uitA*). Approximately 60% bore the $bla_{\text{CTX-M-15}}$ allele, although other versions were also found, including $bla_{\text{CTX-M-55}}$ and $bla_{\text{CTX-M-3}}$. *Escherichia coli* ST44 were found from 2010 onward, except in 2013 and 2017 (Table 1).

ST38 isolates belonged to phylogroup D and were cataloged as ExPEC, with all of them carrying at least two traits (*kpsMII* and *afa*) and some carrying three (*papC*, *kpsMII*, and *afa*). The $bla_{\text{CTX-M-27}}$ variant was the most abundant, followed by $bla_{\text{CTX-M-15/27}}$. In particular, >60% (5/8) showed resistance to FOX, while none showed resistance to GEN. This ST38 was detected from 2016 to 2019 (Table 1). All ST1193 isolates belonged to phylogroup B2 and were classified as ExPEC. The presence of *papA/C*, *kpsMII*, and *uitA* was common in these bacteria. The $bla_{\text{CTX-M-14}}$ allele was the most abundant, although other versions were also registered ($bla_{\text{CTX-M-14}}$, $bla_{\text{CTX-M-15}}$, $bla_{\text{CTX-M-55}}$, $bla_{\text{CTX-M-1/27}}$, $bla_{\text{CTX-M-55}}$, and $bla_{\text{CTX-M-27}}$). All *E. coli* ST88 isolates were associated with phylogroup A and mostly included the $bla_{\text{CTX-M-14}}$ variant. They were classified as non-ExPEC. *Escherichia coli* ST88 was found in 2009, 2010, and 2011

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Table 1. Numbers of total and characteristics of ESBL-producing *E. coli* isolated from bloodstream infection in the Vozandes Hospital, Quito, Ecuador from 2009 to 2019.

Year	Total n° of *E. coli* isolates (*n*,%)	N° of isolates producing ESBLs (*n*; %)	Sequence type (*n*)	Phylogenetic group (*n*)	Type of ESBL (*n*)	Virulence traits (*n*)
2009	53/206; 27.18%	4/53; 7.5%	ST88 (3)	A (3)	CTX-M-15 (3)	uitA (3)
			ST393 (1)	D (1)	CTX-M-14 (1)	iutA, kpsMII, papA, papC, tsh (1)
2010	49/199; 24.62%	7/49; 14.28%	ST131 (1)	B2 (1)	Other CTX-M (1)	afa, iuta, kpsMII (1)
			ST88 (1)	A (1)	CTX-M-15 (1)	uitA (1)
			ST44 (2)	A (2)	CTX-M-15 (1)	iutA, papC (1)
					Other CTX-M (1)	uitA (2)
			ST998 (1)	B2 (1)	CTX-M-15 (1)	afa, iutA, kpsMII, sfa (2)
			ST95 (1)	B2 (1)	CTX-M-22 (1)	iutA, kpsMII, papA, papC (1)
			ST5019 (1)	A (1)	CTX-M-14 (1)	iuta, kpsMII, tsh (1)
2011	33/170; 19.41%	5/33; 15.15%	ST44 (1)	A (1)	CTX-M-15 (1)	uitA (1)
			ST1193 (2)	B2 (2)	CTX-M-14 (2)	iutA, kpsMII (1)
						iutA, kpsMII, papA (1)
			ST88 (1)	A (1)	CTX-M-15/14 (1)	iutA, kpsMII, papC (1)
			ST131 (1)	B2 (1)	CTX-M-15 (1)	iutA, kpsMII, papA, papC (1)
2012	43/167; 25.75%	9/43; 20.93%	ST44 (2)	A (2)	CTX-M-15 (2)	No virulence genes detected (1)
						iutA, papC (1)
			ST131 (4)	B2 (4)	CTX-M-15 (3)	iutA, kpsMII, papA, papC (2)
						iutA, kpsMII, papA, papC, sfa (1)
					Other CTX-M (1)	afa, iutA, kpsMII, sfa (1)
			ST1706 (1)	B1 (1)	CTX-M-22 (1)	uitA (1)
			ST69 (1)	D (1)	CTX-M-22 (1)	iutA, kpsMII, papA, papC (1)
			ST457 (1)	F (1)	CTX-M-103/27 (1)	iutA, kpsMII, papC (1)
2013	40/188; 21.27%	13/40; 32.5%	ST1706 (1)	B1 (1)	CTX-M-15 (1)	iuta, papA, papC (1)
			ST648 (1)	F (1)	CTX-M-15 (1)	iutA, kpsMII, papC, sfa (1)
			ST131 (7)	B2 (7)	CTX-M-15 (7)	iutA, kpsMII, papA, papC (7)
			ST10 (1)	A (1)	CTX-M-14 (1)	afa, iuta, kpsMII (1)
			ST697 (1)	A (1)	CTX-M-15 (1)	uitA (1)
			ST224 (1)	B1 (1)	CTX-M-55 (1)	No virulence genes detected (1)
			ST354 (1)	F (1)	CTX-M-15 (1)	iutA, kpsMII, papC (1)
2014	50/168; 29.76%	20/50; 40%	ST131 (16)	B2 (16)	CTX-M-15 (16)	iutA, kpsMII, papA, papC (16)
			ST44 (1)	A (1)	CTX-M-15 (1)	uitA (1)
			ST1193 (1)	B2 (1)	CTX-M-14 (1)	iuta, papA, papC (1)
			ST2962 (1)	A (1)	CTX-M-15 (1)	No virulence genes detected (1)
			ST453 (1)	B1 (1)	CTX-M-3 (1)	No virulence genes detected (1)
2015	60/161; 37.26%	23/60; 38.33%	ST131 (18)	B2 (18)	CTX-M-15 (16)	iutA, kpsMII, papA, papC (11)
						kpsMII, papA, papC (1)
						afa, kpsMII, uitA, papC (1)
						kpsMII, papC (1)
						iuta, papA, papC (2)
					Other CTX-M (2)	afa, iutA, kpsMII, sfa (1)
						afa, iuta, kpsMII (1)
			ST648 (1)	F (1)	CTX-M-2/97 (1)	iutA, kpsMII, papC (1)
			ST44 (3)	A (3)	CTX-M-15 (2)	No virulence genes detected (1)
						uitA (1)
					CTX-M-3 (1)	uitA (1)
			ST1193 (1)	B2 (1)	CTX-M-55 (1)	iutA, kpsMII (1)
2016	66/157; 42.03%	25/66; 37.87%	ST131 (17)	B2 (17)	CTX-M-15(14)	iutA, kpsMII, papA, papC (11)

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Table 1. Continued

Year	Total n° of *E. coli* isolates (*n*,%)	N° of isolates producing ESBLs (*n*; %)	Sequence type (*n*)	Phylogenetic group (*n*)	Type of ESBL (*n*)	Virulence traits (*n*)
						kpsMII, papA, papC (2)
						iutA, kpsMII, papC (1)
					Other CTX-M (1)	afa, kpsMII, uitA, papC (1)
					CTX-M-170 (1)	iutA, kpsMII, papA, papC (1)
					CTX-M-27 (1)	iutA, kpsMII (1)
			ST457 (1)	F (1)	CTX-M-55 (1)	uitA (1)
			ST38 (2)	D (2)	CTX-M-15/27 (1)	afa, kpsMII, papC (1)
					CTX-M-27(1)	afa, kpsMII (1)
			ST69 (1)	D (1)	CTX-M-3 (1)	iutA, kpsMII, papA, papC (1)
			ST90 (1)	A (1)	CTX-M-55 (1)	iutA, kpsMII (1)
			ST1128 (1)	B1 (1)	CTX-M-55 (1)	uitA (1)
			ST354 (1)	F (1)	CTX-M-2 (1)	uitA (1)
			ST44 (1)	A (1)	CTX-M-55 (1)	uitA (1)
2017	73/163; 44.78%	25/73; 34.24%	ST131 (18)	B2 (18)	CTX-M-15 (16)	iutA, kpsMII, papA, papC (15)
						afa, iuta, kpsMII (1)
					Other CTX-M (1)	afa, iuta, kpsMII (1)
					CTX-M-27 (1)	iutA, kpsMII (1)
			ST10 (2)	A (2)	CTX-M-55 (1)	uitA (1)
					CTX-M-3 (1)	No virulence genes detected (1)
			ST405 (1)	B2 (1)	CTX-M-15 (1)	kpsMII (1)
			ST38 (2)	D (2)	CTX-M-27 (2)	afa, kpsMII (1)
			ST1193 (1)	B2 (1)	CTX-M-27 (1)	iutA, kpsMII (1)
			ST602 (1)	B1 (1)	CTX-M-65 (1)	uitA (1)
2018	43/159; 27.04%	11/43; 25.58%	ST131 (7)	B2 (7)	CTX-M-15 (5)	iutA, kpsMII, papA, papC (5)
					CTX-M-27 (2)	iutA, kpsMII (2)
			ST648 (1)	F (1)	CTX-M-55 (1)	iutA, kpsMII, papC (1)
			ST44 (1)	A (1)	CTX-M-15 (1)	No virulence genes detected (1)
			ST58 (1)	B1 (1)	CTX-M-65 (1)	uitA (1)
			ST38 (1)	D (1)	CTX-M-27 (1)	afa, kpsMII (1)
2019	64/175; 36.57%	24/64; 37.5%	ST131 (14)	B2 (14)	CTX-M-15 (14)	iutA, kpsMII, papA, papC (11)
						iuta, papA, papC (3)
			ST38 (3)	A (3)	CTX-M-27 (2)	afa, kpsMII (1)
						afa, kpsMII, papC (1)
					CTX-M-15/27 (1)	afa, kpsMII, papC (1)
			ST10 (1)	A (1)	CTX-M-8 (1)	papC (1)
			ST44 (2)	A (2)	CTX-M-55 (2)	iutA, papC (1)
						uitA (1)
			ST1193 (2)	B2 (2)	CTX-M-15 (1)	iutA, kpsMII (1)
					CTX-M-1/27 (1)	iutA, kpsMII, papA, papC (1)
			ST5281 (1)	E (1)	CTX-M-55 (1)	iutA, papC (1)
			ST101 (1)	B1 (1)	CTX-M-79 (1)	uitA (1)

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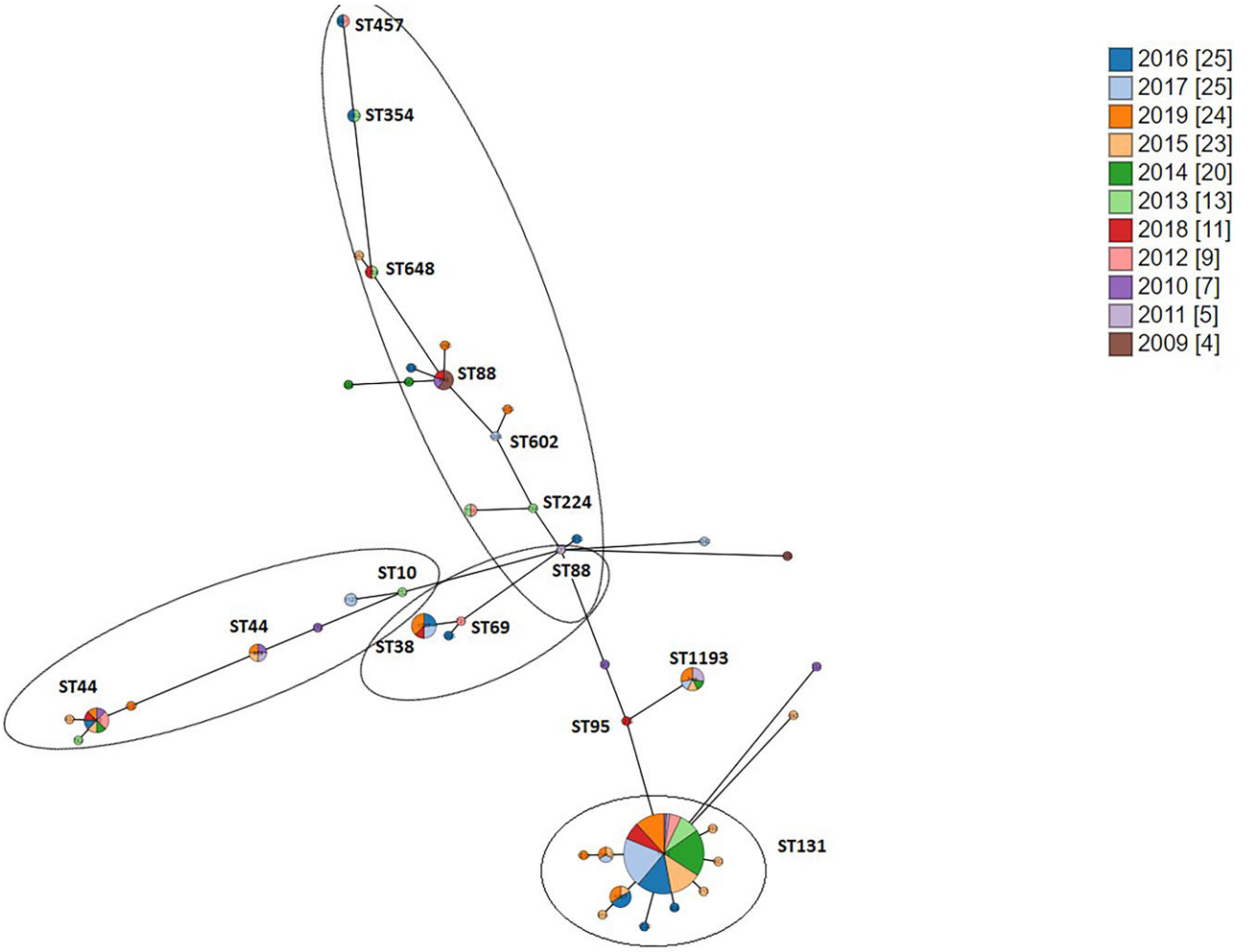


Figure 1. Determination sequence types. Grape trees of isolates were generated using CHtyper and their corresponding specific STs types among isolates.

(Table 1). *Escherichia coli* ST10 belonged to phylogroup A, and only one was categorized as ExPEC because it bore the *kpsMII*, *afa*, and *iutA* traits. The $bla_{\text{CTX-M-14}}$, $bla_{\text{CTX-M-55}}$, $bla_{\text{CTX-M-3}}$, and $bla_{\text{CTX-M-8}}$ variants were found. *Escherichia coli* ST10 was identified in 2009, 2017, and 2019 (Table 1). All the clones described above showed resistance to β-lactams, and fluoroquinolones, and some clones, such as ST10 and ST88, were resistant to GEN.

Discussion

These data provide a distinct and contemporary picture of ESBL-producing *E. coli* BSIs registered in a hospital located in Quito, the capital city of Ecuador, over a period of 10 years (2009–2019). This surveillance study was carried out at a time when the incidence of *E. coli* BSI had been classified as important in the region (Zurita et al. 2019, Rosenthal et al. 2021), especially with the emergence of multidrug resistant strains (Salles et al. 2013).

Latin America has seen a steady increase in the prevalence and types of extended-spectrum β-lactamases (particularly $bla_{\text{CTX-M}}$) produced by *E. coli* isolates in Latin American hospitals. According to a multicenter study, the rates of nosocomial infections caused by *E. coli* that produce extended-spectrum β-lactamases increased by 32% from 2005 to 2010. These are the highest rates compared to other regions of the world (Guzmán-Blanco et al. 2014). At the country level, the highest resistance rates were observed in Chile, Ecuador, Guatemala, Honduras, and Mexico. Additionally, the resistance of extended-spectrum β-lactamase-producing Enterobacteriaceae to FEP, fluoroquinolones, SAM, aminoglycosides, and TPZ has increased from 2005 to 2010. Carbapenems, TGC, and colistin (COL) are the only antibiotics with susceptibility rates >90% (Guzmán-Blanco et al. 2014). Similar results were obtained in our study, with the exception of carbapenems and TGC; there were no resistant isolates during these years, and TPZ did not show any significant increase.

A previous study demonstrated the importance of ESBL-producing *E.coli* in BSI in Ecuador (Zurita et al. 2019). The $bla_{\text{CTX-M-15}}$ variant, which was the most common, encodes a β-lactamase that hydrolyzes third-generation cephalosporins, such as cefotaxime and CRO (Tzouvelekis et al. 2000), which are widely used in Ecuador. This gene has been reported in studies carried out in the area, although it is poorly represented in the nonhuman context (Ortega-Paredes et al. 2018a, 2018b; Zurita et al. 2019). This variant has been observed in ready-to-eat food, which is probably caused by cross-contamination during preparation, as well as in urban river samples associated with anthropogenic wastewater and in dogs kept as pets (Ortega-Paredes et al. 2019,

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2020; Albán et al. 2020). Other variants conferring resistance to cephalosporins used in Ecuador were also registered, including $bla_{\text{CTX-M-27}}$, $bla_{\text{CTX-M-55}}$, and $bla_{\text{CTX-M-14}}$. These genes have been documented not only in BSI (Zurita et al. 2019) but also in animal and environmental samples collected in Quito (Vinueza-Burgos et al. 2019, Zurita et al. 2020).

The majority of isolates were resistant to SAM, GEN, CRO, and CIP. Overall, the number of resistant bacteria increased over time. Some isolates were not only resistant to the aforementioned drugs but also resistant to FOX, CAZ, and FEP. Seven ESBL-producing *E. coli* isolates, which were considered resistant to CRO according to the CLSI breakpoints, were categorized as susceptible by the VITEK®2 system. The observed differences could thus be linked to inconsistencies of the system (Bobenchik et al. 2015) or factors not established (Tamma and Humphries 2021).

Various isolates showed resistance to FOX. This antibiotic is non-marketed in Ecuador; therefore, this particularity may be a result of indirect selection due to the use of third-generation cephalosporins. In fact, the $bla_{\text{CMY-2}}$ gene has been reported in *E. coli* populations obtained from various sources (Ortega-Paredes et al. 2019, Vinueza-Burgos et al. 2019). The $bla_{\text{CMY-2}}$-encoded enzyme has a hydrolysis spectrum that includes third-generation cephalosporins, such as cefotaxime and CRO, which are widely used in the human context, as well as ceftiofur, a third-generation cephalosporin that is widely applied in veterinary medicine. It is advisable to perform in-depth studies of the dynamics of the $bla_{\text{CTX-M}}$ and bla_{CMY} genes as a function of the third-generation cephalosporins that exert selective pressure. The studied ESBL-producing *E. coli* showed susceptibility to IPM, MEM, TGC, and ETP. The absence of these carbapenemases in BSI in this hospital, could be related to the hospital infection control program, which includes patient bathing, hand washing, compliance with isolation standards, and a stewardship program in recent years (Quirós et al. 2022).

Most isolates were classified as ExPEC, and the number of isolates tended to increase over time. Infections by extraintestinal pathogenic *E. coli* have been described in Ecuador, especially UTIs and BSIs caused by ESBL-producing *E. coli* (Calva et al. 2016, Chiluisa-Guacho et al. 2018, Soria Segarra et al. 2018, Zurita et al. 2019). Most isolates harbored genes encoding structural (*pap*A) and assembly proteins (*pap*C) of P fimbriae, and they also carried the *kpsMII* and *uitA* genes involved in Group 2 capsule synthesis and iron uptake, respectively. Thus, these ESBL-producing bacteria appeared capable of not only binding to epithelial cells via P fimbria (Pap proteins) but were also most likely to resist neutrophil phagocytosis and complement-mediated bactericidal activity, resulting in enhanced bloodstream survival linked with the presence of the bacterial capsular antigen K. In addition, they would be capable of iron acquisition mediated by siderophores (*uitA*) (Schwan et al. 2005, Volkan et al. 2012, Robinson et al. 2018). Our results showed that *iutA* and *kpsMII* are present in almost all isolates. Undoubtedly, further research should emphasize the study of these genes because they might play a crucial role in bacteremia associated with extraintestinal infections and could even be used as potential predictors of bacterial invasion.

ST131 was the most abundant sequence type, and related isolates were mostly classified as ExPEC and bore mainly the $bla_{\text{CTX-M-15}}$ variant. These isolates were susceptible to β-lactam antibiotics and were resistant to GEN and CIP. ST131 is regarded as a major cause of extraintestinal infections worldwide (Nicolas-Chanoine et al. 2014). Additionally, the resistance patterns identified in the ST131 strains suggest the circulation of sub-lineages, such as H30R, which is characterized by its resistance to CIP; H30RxC1, which is characterized by its resistance to CIP and ESBL; and H30RxC2, which is characterized by its resistance to GEN and CIP and ESBL. Clade C1-M27 is associated with $bla_{\text{CTX-M-27}}$, and C2 is associated with $bla_{\text{CTX-M-15}}$ (Peirano and Pitout 2019). Other studies have documented ST131 strains with similar characteristics from BSI and environmental samples (Ghosh et al. 2017, Merino et al. 2018). Humans are considered the reservoir of this ST131. Therefore, the fecal-oral route must have played a major role in the transmission of this sequence type (Sarowska et al. 2019). The latter ST was associated with $bla_{\text{CTX-M-15}}$, although $bla_{\text{CTX-M-55}}$ and $bla_{\text{CTX-M-3}}$ were also registered and have been identified in environmental samples in Quito (Ortega-Paredes et al. 2020, Zurita et al. 2020). *E. coli* ST10 was not associated with $bla_{\text{CTX-M-15}}$ but rather was associated with $bla_{\text{CTX-M-14}}$, $bla_{\text{CTX-M-55}}$, and $bla_{\text{CTX-M-3}}$, and these genes have been commonly registered in studies analyzing environmental samples gathered in Quito (Ortega-Paredes et al. 2020, Zurita et al. 2020). ST88 appeared to be the origin of various clones and it was predominantly related to $bla_{\text{CTX-M-14}}$. *E. coli* ST88 has been detected in environmental studies as well as in BSI (Borges et al. 2019).

This surveillance study of ESBL-producing *E. coli* from BSI shows the dominance of ST131, B2, $bla_{\text{CTX-M-15}}$, and *pap*A-*pap*C-*kps*MII-*uit*A. Additionally, these data support the increase in the ExPEC ESBL-*E coli* problem in BSI and highlight the relationship between the isolates found in these infections and those that are circulating in the environment. Thus, surveillance of these strains should not be limited to hospitals and should be extended to the community and environment.

Study limitations

Since the study was conducted at a single private center, the results may not be applicable in settings with a different epidemiological context. Information on risk factors, such as previous antibiotic exposure, sources of bacteraemia and transfer from other facilities, was not included in our analyses.

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Supplementary data

Supplementary data is available at *JAMBIO Journal* online.

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Author contributions

Jeannete Zurita (Conceptualization, Formal analysis, Investigation, Project administration, Supervision, Writing – original draft, Writing – review & editing), Gabriela Sevillano (Data curation, Formal analysis, Methodology, Writing – original draft), Ariane Paz y Miño (Formal analysis, Methodology, Writing – original draft), Nathalí Haro (Formal analysis, Investigation, Methodology), Marco Larrea-Álvarez (Formal analysis, Writing – original draft), Iliana Alcocer (Project administration, Resources), and David Ortega-Paredes (Conceptualization, Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing)

Data availability

The data that supports the findings of this study are available in the supplementary material of this article.

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